FRA Operations Forecast - model card

30 October 2026 | Hackathon research demo | Partial challenge completion

Purpose and outputs

48 hourly forecasts of Frankfurt wind and provisional RWY18 weather usability. The separate Section 3 baseline predicts airport-wide 07/25 and change windows. Learned actual OD, certainty, and change accuracy remain unavailable without independent historical labels. Weather suitability is not official runway status or actual use.

Model and key drivers

Validation selected Ridge residual correction (alpha=1) over raw ICON-EU, boosting, and a fixed equal blend; observation persistence is a reference. Inputs: raw u/v wind, gust and missing indicator, model lead, UTC hour and season. Train-only imputation/scaling; no importance ranking claimed. RWY18 probabilities resample 256 joint residual draws from separate calibration pools. The optional AI coordinator explains the frozen selection; it cannot change numeric weights.

Sources and time design

Open-Meteo explicit ICON-EU runs supply forecast inputs; DWD station 01420 supplies 10-minute wind and preceding-hour maximum gust targets. Live context: AWC EDDF METAR and UNH/DFS current OD. GFS via Open-Meteo passed four 96-hour probes but has no ranked historical evaluation. Source retrieval metadata and hashes are saved.

Issue-date partitions (2026): train Apr 30-Aug 15; selection Aug 16-Sep 02; calibration Sep 03-20; test Sep 21-Oct 26, targets through Oct 29. Earlier targets crossing the next partition were purged (1,224 rows). Selection used 3,248 matched 1-48 h cases. Historical forecast availability assumes four hours for ICON-EU, six for GFS; observations assume one hour. Publication history and later DWD corrections are unverified.

Measured quality - lower scores are better

Validation vector RMSE: raw 5.517 -> selected 4.377 kt; gust MAE: 3.751 -> 3.019 kt. Below: previously inspected test period, explicitly exploratory; 144 cases per horizon. Wind errors are knots. RWY18 Brier is compared with Section 3 on the same cases.

Lead	Raw vector RMSE	Selected vector RMSE	Selected gust MAE	Selected RWY18 Brier	Section 3 Brier
1 h	4.232	2.951	2.473	0.000016	0.000000
6 h	4.489	3.005	2.704	0.006957	0.006944
24 h	4.518	3.209	2.680	0.006997	0.006944

Across 1-48 h: selected Brier 0.001195 vs always-usable 0.001157 (6,912 cases); matched Section 3 comparison 0.001197 vs 0.001159 (6,902 cases). Eight adverse forecast cases represent only ONE distinct adverse timestamp; all were missed at probability threshold 0.5, with zero false warnings. No reliable rare-event calibration or operational baseline improvement is established. Actual-OD accuracy and change timing cannot yet be scored.

Limits and delivery status

Probabilities: provisional_low_event_coverage. Hourly inputs cannot resolve exact >30-minute persistence or within-hour reversals; missing gusts stay missing. Correction candidates share ICON-EU inputs and are correlated. Season/generalization and untouched holdout performance remain unverified. The brief does not supply jury dates, issue times, or file schema. Replay/export works: an internal demo issue produced 48 weather rows, null learned OD, and an explicit incomplete report. Agent provider execution requires configuration and is unverified. This is not an operational aviation system.

Reproducibility

Rules: fraalliance-2026-v1 (landing 10 kt, departure 15 kt, crosswind >40 kt, persistence >30 min). Model: wind-comparison-v1; experiment ed131c7b15d7291b15a5; scikit-learn 1.9.1. Dataset SHA-256: 66fe17ae25c62ea2fda4031a678e4bfe379f27a268ea98dc4dfd56098aa2de1b. Frozen artifacts, source metadata, reports and dataset are in the local demo bundle; commands and full identities are in docs/model-comparison.md. Model fitting/selection/calibration cutoff: 2026-09-21 00:00 UTC.